Supplementary Information for
A Hopfield attractor account of numerosity perception

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This document accompanies the article *A Hopfield attractor account of numerosity perception*.

Supplementary Material 1: Gaussian wells with Benford-weighted depth

The Poisson parameterization presented in the main text is motivated by the accumulator statistics of numerosity perception. A natural alternative starts from a different premise: that integers are stored memories whose strength reflects environmental exposure. The frequency with which number words appear in natural language follows a distribution closely approximating Benford's law across typologically diverse languages (Dehaene and Mehler 1992), with small numbers occurring far more frequently than large ones. If the attractor wells are Gaussian and their width is shaped by this frequency, then small, frequent numbers occupy narrower and more stable basins, while larger, rarer numbers occupy broader and more overlapping basins (Fig. S1).

This framing connects the attractor model to the broader view of numbers as memory traces whose strength can be shaped by experience (Anderson and Schooler 1991; Verguts and Fias 2004). It is not the model used for the main analyses, but it is useful as a conceptual comparison. In the Poisson model, the broadening of wells follows directly from accumulation statistics: $\mathrm{Var}(x \mid n) = cn$. In the Gaussian–Benford variant, the broadening is driven instead by the environmental frequency of each integer.

Model specification

Each integer n is represented by a Gaussian attractor well centred at $\mu_n = n$, with width determined by the Benford frequency $f_n = \log_{10}(1 + 1/n)$:

$$\sigma_n = \left(\frac{c_G}{f_n}\right)^{\beta}, \tag{S1}$$

where c_G is a base width scale and β controls how strongly frequency modulates well width ($\beta = 1$: linear; $\beta < 1$: attenuated). The well potential for each integer is:

$$g_n(x) = -\log \sigma_n - \frac{(x-n)^2}{2\,\sigma_n^2}. \tag{S2}$$

The full attraction potential retains the same Hopfield structure as in the main text (Eq. 2), with the Gaussian

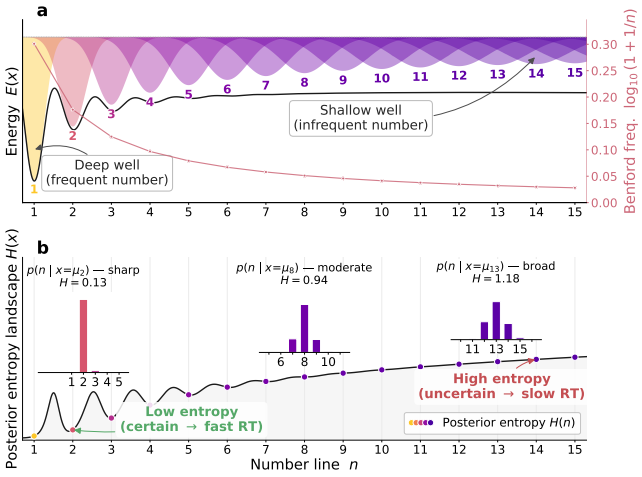


Figure S1. Energy landscape of the Gaussian–Benford attractor wells for $n = 1 \ldots 15$. In **a**, the energy landscape $E(x)$ is shaped by Gaussian wells whose width is modulated by Benford's law frequency (pink curve, right axis): $n = 1$, the most frequent number, occupies the narrowest basin, while larger, rarer numbers produce progressively wider and more overlapping basins. This contrasts with the Poisson parameterization (main text, Fig. 1), where the same qualitative asymmetry arises from accumulator statistics rather than environmental exposure. In **b**, the posterior entropy landscape $H(n)$ illustrates the consequences for pattern completion: small numerosities yield sharply peaked posteriors, whereas larger numerosities spread probability over several neighbouring wells.

g_n replacing the Poisson form. The posterior readout is also analogous to the main model, except that the Benford frequency can enter as a prior:

$$p(n \mid x^*) \propto f_n \cdot \exp(g_n(x^*)). \tag{S3}$$

The sensory evidence channel and response-time model are otherwise the same as in the main text. Under the final parameterization used in the article, the comparable free parameters would be c_G (Gaussian well scale), s_0 and λ_0

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Table S1. Summary-level model comparison on the Cheyette and Piantadosi (2020) duration data. AIC/BIC are profile-Gaussian values averaged across the five exposure durations, computed on mean estimate, exact accuracy, and median RT. Δ is the difference in mean AIC from the best model and w the corresponding Akaike weight. LONO is leave-one-numerosity-out normalized predictive loss; LODO is leave-one-duration-out loss. Lower values are better.

Model	k	AIC	Δ	w	BIC	LONO	LODO
Hopfield–Poisson	**4**	**106.9**	**0.0**	**0.52**	**114.2**	**0.007**	**0.010**
No-cleanup Poisson	4	108.5	1.5	0.25	115.7	0.007	0.013
Gaussian/Benford readout	4	109.4	2.4	0.16	116.6	0.007	0.010
Weber/ANS baseline	4	110.7	3.8	0.08	118.0	0.008	0.015
C&P log-prior proxy	4	121.9	15.0	0.00	129.1	0.011	0.015

Table S2. Per-duration AIC and Akaike weights. Profile-Gaussian AIC for each model and exposure duration; Akaike weights (in parentheses) are computed within each duration. Bold marks the lowest AIC in each column.

Model	40 ms	80 ms	160 ms	320 ms	640 ms
Hopfield–Poisson	131.8 (.01)	**82.6** (.75)	**81.4** (.52)	118.6 (.06)	120.3 (.32)
No-cleanup Poisson	130.4 (.02)	88.9 (.03)	85.4 (.07)	**113.1** (.93)	124.5 (.04)
Gaussian/Benford	130.8 (.02)	85.6 (.17)	83.6 (.17)	127.9 (.00)	**118.9** (.64)
Weber/ANS	125.2 (.26)	88.1 (.05)	83.0 (.24)	124.2 (.00)	133.2 (.00)
C&P log-prior proxy	**123.3** (.69)	98.8 (.00)	101.6 (.00)	137.0 (.00)	149.0 (.00)

Table S3. Fitted parameters across exposure durations.

Duration	c	s_0	λ_0	κ
40 ms	0.079 [0.069,0.090]	0.085 [0.053,0.100]	1.98 [1.67,2.35]	0.72 [0.66,0.77]
80 ms	0.049 [0.045,0.053]	0.0069 [0.0048,0.0098]	2.99 [2.80,3.19]	0.88 [0.83,0.93]
160 ms	0.043 [0.040,0.047]	0.0018 [0.0014,0.0022]	3.40 [3.28,3.52]	0.97 [0.91,1.03]
320 ms	0.039 [0.035,0.045]	0.00068 [0.00053,0.00088]	3.69 [3.59,3.80]	1.01 [0.92,1.12]
640 ms	0.035 [0.029,0.041]	0.00030 [0.00024,0.00038]	3.86 [3.78,3.95]	1.06 [0.93,1.22]

(evidence noise), and κ (entropy-to-RT scaling), with the basal response time t_0 fixed by task.

Comparison and interpretation

The Gaussian–Benford construction produces the same qualitative geometry that motivates the main model: narrow, well-separated basins for small numerosities and broader, overlapping basins for larger ones. It therefore provides an alternative route to functional compression of the number line. The interpretation is different, however. In the Poisson model, compression follows from the statistics of accumulating n pieces of evidence. In the Gaussian–Benford variant, compression follows from an assumed frequency-dependent memory strength.

For the present article we focus on the Poisson parameterization because it is the more direct expression of the accumulator hypothesis and because it requires only one well-scale parameter, c, to determine the full family of attractor widths. The Gaussian–Benford variant remains useful as a conceptual comparison and suggests a possible extension: exposure frequency could modulate the depth or precision of wells on top of the accumulator-derived Poisson geometry. This hybrid possibility may be especially relevant for developmental data, where exposure history changes systematically with age, but testing it would require additional data and is outside the scope of the present first evaluation.

Supplementary Material 2: Model comparison and functional compression

The main text reports the AIC comparison across exposure durations. Table S1 gives the corresponding summary-level information criteria, Akaike weights, and predictive losses. The C&P row is a conservative log-prior proxy fitted to the same summary curves, not a full reproduction of the original trial-level Bayesian observer. The Akaike weight w is computed from the across-duration mean AIC and gives the relative probability that each model is the best in this set (Burnham and Anderson 2002).

The separation among the three Poisson/Gaussian-family models is modest. The Hopfield–Poisson model carries about half of the total Akaike weight, and the no-clean-up control that removes the attractor step is only $\Delta\text{AIC} = 1.5$ behind; on the summary curves alone the data do not decisively prefer one over the other. The advantage is also unevenly distributed across exposures (Table S2): the Hopfield–Poisson model has the lowest AIC at 80 and 160 ms but is matched or beaten at 40, 320 and 640 ms, and it improves on the no-clean-up control in three of the five durations. As argued in the main text, the case for the attractor therefore rests on its mechanistic and out-of-sample-predictive contributions (entropy-based RT, the n_{crit} prediction, and lower LODO loss) rather than on a decisive in-sample fit advantage.

Table S3 gives the fitted parameters for the Hopfield–Poisson model across exposure durations. Point estimates minimize the three-channel goodness-of-fit loss used in the main text; bracketed ranges are 95% log-scale Wald confidence intervals from the profile-Gaussian Hessian. The task-specific basal response time is fixed at $t_0 = 1.30$ s.

Is the duration dependence a change of mechanism? A shared-landscape control

A reasonable concern about fitting each exposure independently is whether it implies that the perceptual mechanism itself differs across timing conditions. To address this we refit the data with the two attractor/readout parameters held common across durations — a single well scale c and a single entropy-to-RT coefficient κ — while allowing only the sensory-noise parameters s_0 and λ_0 to vary per duration. This shared-landscape model has $2 + 2 \times 5 = 12$ parameters against the $4 \times 5 = 20$ of the per-duration fit. We compare the two using pooled profile-Gaussian information criteria over all five durations (Table S4).

The constrained model is clearly rejected in both poolings ($\Delta\text{AIC} = 49$ across all five durations and 79 across the longer four; BIC agrees), so the per-duration variation in c and κ is statistically real and not an artefact of overfitting. This does not, however, indicate that the mechanism changes with timing. The shared-landscape model still reproduces the longer-duration curves well in absolute terms ($R^2 = 0.87$–0.97 for the 80–640 ms conditions), so the rejection reflects

Table S4. Shared-landscape control. Pooled profile-Gaussian information criteria for the per-duration fit (free c, κ) versus the constrained model with c and κ shared across durations and only s_0, λ_0 free. "All" pools the five durations; "80–640 ms" pools the four longer, more informative durations. Lower is better.

Durations	Model	k	AIC	BIC
All (5)	Per-duration (free c, κ)	20	**533.1**	**601.4**
	Shared c & κ	12	582.3	623.3
80–640 (4)	Per-duration (free c, κ)	16	**401.4**	**452.5**
	Shared c & κ	10	480.1	512.1

the high precision with which the per-duration model fits these data rather than a qualitative failure. More importantly, the way c and κ move is smooth and monotonic in log-exposure (Table S3): c decreases from 0.079 at 40 ms to 0.035 at 640 ms as the input becomes informative enough to constrain the landscape, and κ increases from 0.72 to 1.06 as response time becomes more sensitive to residual posterior uncertainty. These graded, interpretable trends are exactly what the leave-one-duration-out analysis exploits to predict held-out exposures, and they are consistent with a single mechanism receiving progressively better input and using a slightly higher readout gain, rather than a reorganization of the perceptual process across timing conditions.

Supplementary Material 3: Parameter Recovery

To assess whether the fitted parameters were identifiable from the summary-level data used in the main analyses, we ran a parametric parameter-recovery analysis across all five exposure durations from Cheyette and Piantadosi (2020). For each duration, synthetic datasets were generated from jittered parameter values around the fitted solution and refitted using the same optimization pipeline as in the main text. The recovery sweep used 50 synthetic datasets per duration, yielding 250 refits in total. Each synthetic dataset contained the same numerosity range as the empirical analysis, and the recovered parameters were compared with the generating values on the original parameter scales.

Figure S2 shows the recovered parameters against the true generating values. The attractor well-scale parameter c, which controls the effective width of the Poisson wells and therefore adjacent-integer separability, was recovered accurately across durations ($R^2 = .95$, Spearman $\rho = .96$). The entropy-to-response-time scaling parameter κ was also recovered accurately ($R^2 = .87$, $\rho = .96$). The sensory-noise parameters were less cleanly separable. The noise scale s_0 showed high rank-order recovery but some systematic bias ($R^2 = .92$, $\rho = .82$), while the exponent λ_0 showed weaker linear recovery despite good rank-order recovery ($R^2 = .49$, $\rho = .83$). This pattern indicates that s_0 and λ_0 should not be interpreted as independent psychological quantities. Instead, they should be interpreted jointly as defining the sensory-noise profile $\sigma(n) = \sqrt{s_0 n^{\lambda_0}}$.

The error-correlation matrix in Figure S2 confirms this interpretation. Recovery errors for s_0 and λ_0 were negatively correlated ($\rho = -.68$), indicating a trade-off between the two parameters: increases in the fitted noise scale can be partly compensated by decreases in the fitted exponent, and

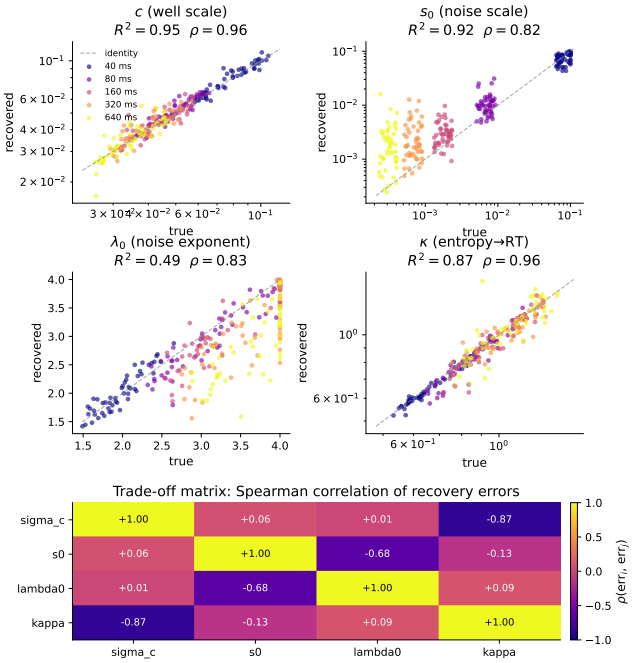


Figure S2. Parameter-recovery analysis across the five exposure durations. Each point corresponds to one synthetic dataset generated from jittered fitted parameters and then refitted with the same optimization procedure used in the main analyses. Colours indicate exposure duration. The dashed grey lines show identity. The attractor well-scale parameter c and the entropy-to-response-time scaling parameter κ are recovered accurately. The sensory-noise parameters s_0 and λ_0 show partial trade-off, indicating that they should be interpreted jointly as defining the sensory-noise profile rather than as separately identifiable psychological quantities. The bottom panel shows the Spearman correlation matrix of recovery errors, highlighting the main trade-offs between s_0 and λ_0, and between c and κ.

vice versa. A second trade-off appeared between c and κ ($\rho = -.87$), consistent with the fact that broader attractor wells increase posterior entropy, which can be partially compensated by a smaller entropy-to-RT scaling coefficient. Importantly, this trade-off did not prevent accurate recovery of either parameter individually. Thus, the recovery analysis supports the main interpretation of c as a recoverable landscape-level parameter, while motivating a more cautious joint interpretation of the two sensory-noise parameters.

Supplementary Material 4: Compression of the number line and transition to density regime

The Poisson attractor landscape also produces functional compression of the number line (Fig. S3). Although well centres are equispaced, the posterior distributions broaden with numerosity, reducing discriminability between adjacent integers.

This compression carries on until an analytic transition point $n_{\mathrm{crit}} = 1/c$ at which adjacent attractor wells stop being reliably distinguishable. With the fitted values of c, this gives $n_{\mathrm{crit}} \approx 13$ in the 40 ms condition and $n_{\mathrm{crit}} \approx 29$ in the 640 ms condition. This section asks how these values compare to empirical estimates of where individual-number perception gives way to texture- or density-based

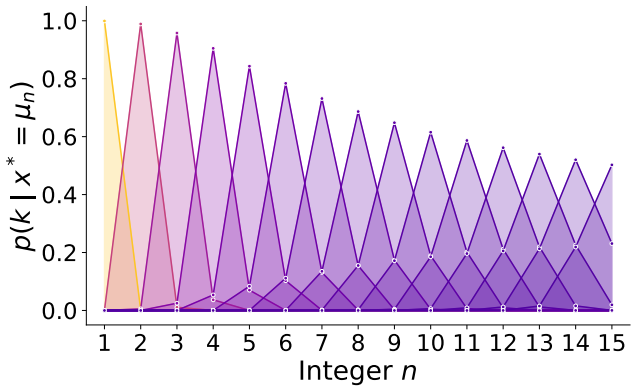


Figure S3. Functional compression from Poisson attractor wells. Posterior distributions at well centres: for small n the posteriors are sharply peaked and non-overlapping, supporting exact identification; as n grows the distributions broaden and merge, producing the functional compression of the number line characteristic of approximate numerosity.

perception in the psychophysics literature. The comparison is necessarily indirect, for reasons discussed below; we therefore report it as a qualitative consistency check rather than a quantitative test of the model.

Computing density from the Cheyette and Piantadosi (2020) display. Stimuli were presented within a circular region of radius 200 pixels, giving a display area of approximately 1.26×10^5 pixels^2. At $n_{\text{crit}} = 29$, this corresponds to a numerical density of 2.3×10^{-4} dots/pixel^2, or about 4 300 pixels per dot. Translating this into degrees of visual angle requires the pixels-per-degree of the participants' displays, which were not controlled: the experiment was conducted online with browser windows ranging from 820×524 to 2560×1349 pixels, and viewing distance was unknown. Plausible web-viewing setups (laptop or desktop displays at typical reading distances) produce roughly 30–50 pixels per degree of visual angle, which would place the 200-pixel-radius display in the range of approximately 50–140 deg^2 and the $n = 29$ threshold at approximately 0.2–0.6 dots/deg^2. The corresponding range for $n = 13$ (40 ms condition) is approximately 0.1–0.3 dots/deg^2.

Comparison with Anobile et al. (2014). Anobile et al. (2014) reported a slope change in numerosity discrimination at 29 dots (0.2 dots/deg^2) for a large reference patch and 14 dots (0.3 dots/deg^2) for a small reference patch. The numerical thresholds (29 and 14 dots) coincide closely with the model's two extreme predictions (13 and 29 dots), and the reported densities lie within the range estimated above for the C&P display under typical web-viewing assumptions.

Comparison with Anobile et al. (2015). Anobile et al. (2015) reported substantially higher density thresholds: the regime change occurred at 2.3 dots/deg^2 in central vision and 0.5 dots/deg^2 at 15° eccentricity. These values are an order of magnitude above the 2014 estimates and above the range derived above for the C&P display. The 2015 paper also documented strong dependence on stimulus eccentricity, patch size, and crowding-like factors, and the authors themselves frame the boundary as governed by spatial-window properties of the visual system rather than as a single fixed numerosity. In the present model, c is fitted from behavioural data and incorporates whatever upstream encoding the participants used; it is therefore not surprising that c varies with display conditions and that a single n_{crit} cannot match all empirical estimates simultaneously.

Interpretation. The agreement with Anobile et al. (2014) is closer than the agreement with Anobile et al. (2015), and the differences between the two empirical estimates are themselves comparable to the differences between either estimate and the model. Because the C&P display geometry is not fully recoverable, none of these comparisons constitutes a quantitative test of the model's prediction. What the comparison does establish is that $n_{\text{crit}} = 1/c$, derived analytically from the fitted attractor scale and not from any density data, falls within the order-of-magnitude range of independently published density-regime thresholds. A direct test would require an experiment in which the same participants provide both a numerosity-estimation dataset (for fitting c) and a density-discrimination dataset (for measuring the empirical regime boundary) in a single controlled visual environment.

Supplementary Material 5: Exploratory reanalysis of developmental dyscalculia data

As an exploratory extension, we applied the same Hopfield–Poisson fitting framework to the exact enumeration data from Decarli et al. (2020). This analysis should be read as hypothesis-generating rather than confirmatory. The dataset contains one sample of children with developmental dyscalculia (DD, $n = 32$) and age-matched controls ($n = 32$), measured on a single exact-enumeration task with unlimited viewing time. Because the displays were not brief, children could have counted or used other task-specific strategies, so the fitted parameters cannot by themselves identify a purely perceptual or representational locus.

We fitted the free parameters separately for DD and control groups using exact accuracy and response-time curves. The task-specific basal response time was fixed to $t_0 = 0.36$ s. The fit used numerosities $n = 1$–7, excluding $n = 8$ because the empirical response pattern at that numerosity was anomalous. This exclusion follows the current fitting criterion, but it should not be treated as analytically innocuous: because it affects a small dataset, it remains a researcher degree of freedom and contributes to the exploratory status of the analysis.

The model captures the broad group pattern but not all details of the accuracy curves (Fig. S4a–b). Exact accuracy is fitted moderately well ($R^2 = 0.85$ for DD and 0.83 for controls; RMSE ≈ 10 percentage points), whereas the response-time profiles are captured more closely ($R^2 = 0.96$ and 0.98, respectively). Panel c reports the fitted parameter summaries separately for each group, using them only as descriptive model quantities.

Within the model, the fitted parameters describe the precision of the attractor landscape, the sensory evidence channel, and the entropy-to-response-time mapping (Fig. S4c). These parameters should not be read individually, however, because they trade off during fitting (Supplementary Material 3): the well scale c and the entropy-to-RT coefficient κ are negatively correlated, as are the two sensory-noise parameters

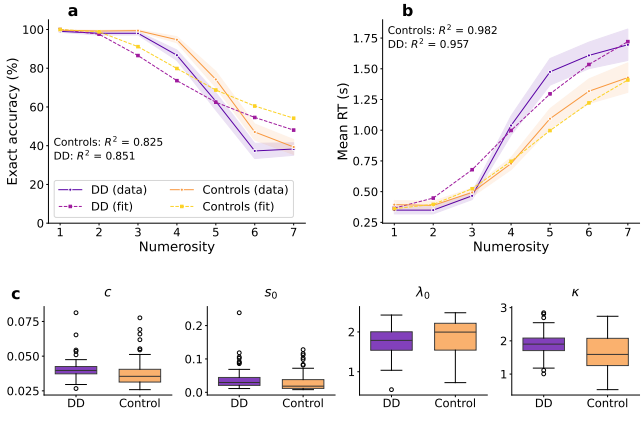


Figure S4. Exploratory dyscalculia reanalysis. (a–b) Exact enumeration accuracy (a) and reaction time (b) from Decarli et al. (2020) for children with developmental dyscalculia (DD) and age-matched controls, with model fits overlaid. The analysis uses numerosities $n = 1$–7; $n = 8$ was excluded because of an anomalous empirical response pattern. (c) Fitted model-parameter summaries shown separately for each group; box plots summarize bootstrap or profile-likelihood draws. The panel is descriptive.

s_0 and λ_0. A group comparison phrased in terms of single parameters would therefore overstate what the data identify. To respect this, we compare the two groups at the level of the recoverable derived quantities that the model is actually built to express: the breakdown point of the discrete code, $n_{\text{crit}} = 1/c$, and the sensory-noise profile $\sigma(n) = \sqrt{s_0\, n^{\lambda_0}}$ summarized over the fitted range. The two are then contrasted using the per-group bootstrap draws.

Read this way, the picture is more cautious than a parameter-by-parameter comparison would suggest. The attractor-precision quantity barely separates the groups: n_{crit} is 25.2 $[18.5, 31.1]$ for DD and 28.2 $[15.6, 36.8]$ for controls (median and 95% bootstrap interval), a difference whose interval comfortably spans zero. The more consistent group difference is in the sensory channel: the mean noise SD over $n = 1$–7 is higher in DD than in controls in 83% of paired bootstrap draws (median difference $+0.06$, interval $[-0.07, 0.14]$), with the same direction at each individual numerosity. The model-based description of the data is thus one of a somewhat noisier sensory input in DD, with at best weak and not separately identifiable evidence for a coarser attractor landscape. This is consistent in direction with neurocomputational accounts in which dyscalculia reflects reduced representational precision of numerosity rather than stronger interference from non-numerical magnitudes (Dolfi et al. 2024), although the present data cannot adjudicate the locus of that reduction.

This exploratory reanalysis may still be useful for future work because it suggests measurable model quantities that could be tested in larger, task-diverse samples. Any implications for intervention remain speculative: the present dataset does not evaluate training, causal mechanisms, or whether an intervention would change attractor-well precision rather than upstream encoding or response strategies.

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